

HURON LIGHT, OH, USA

WMO: 998001

Lat: 41.4005N

Lon: 82.5452W

Elev: 184

StdP: 99.13

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.6	(c) -13.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.2	(k) -4.3	(l) 12.6	(m) -4.5	(n) 4.7	(o) 230	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.8	(c) 31.4	(d) N/A	(e) 29.8	(f) N/A	(g) 28.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.8	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.5	(b) 10.9	(c) 9.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-3.3	-1.7	2.7	9.8	16.2	21.2	23.3	22.1	19.2	12.6	6.1	1.4		
	DBStd	10.38	6.56	6.07	5.47	4.92	4.64	3.34	2.97	2.37	3.74	4.22	5.13	5.27		
	HDD10.0	1478	416	331	237	63	3	0	0	0	0	19	139	269		
	HDD18.3	3198	672	562	483	261	100	11	1	2	33	183	368	523		
	CDD10.0	1796	2	2	12	57	194	334	414	376	276	102	21	4		
	CDD18.3	474	0	0	0	5	33	95	156	119	58	7	0	0		
	CDH23.3	2871	0	0	1	52	213	544	1125	587	322	28	0	0		
	CDH26.7	713	0	0	0	8	54	148	301	106	93	3	0	0		
(14) Wind	WSAvg	4.6	5.3	4.9	5.0	5.3	4.4	4.2	3.8	3.8	4.2	4.8	5.0	4.9		
(15) Precipitation	PrecAvg	916	58	53	62	93	89	105	94	81	85	65	69	66		
	PrecMax	1128	132	125	119	142	162	196	202	198	196	99	141	168		
	PrecMin	637	16	12	26	46	21	40	34	11	26	17	17	24		
	PrecStd	139	29	29	26	25	41	40	43	48	43	21	26	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.5	21.2	28.0	30.8	32.6	33.9	31.6	32.3	25.8	20.4	17.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	13.2	13.3	17.7	24.7	28.1	30.3	31.5	29.4	29.6	23.7	17.9	14.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	9.1	9.7	15.2	21.8	25.7	28.3	29.9	27.7	26.6	21.3	16.0	12.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	5.4	6.2	11.8	18.7	23.5	26.6	28.2	26.4	24.8	19.1	14.1	9.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	6.9	6.6	7.9	7.3	6.6	6.8	6.5	6.8	7.2	6.6	6.0		
		MCDBR	10.5	11.4	11.6	13.7	11.3	9.2	9.5	8.8	9.8	10.3	9.6	9.7		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.335	0.359	0.399	0.433	0.449	0.446	0.433	0.387	0.367	0.329	0.315		
	taud		2.401	2.298	2.310	2.256	2.203	2.206	2.236	2.258	2.361	2.396	2.464	2.435		
	Ebn at Noon		841	865	886	872	847	832	829	829	846	821	811	801		
	Edh at Noon		81	104	116	131	142	142	137	130	109	94	76	73		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.22	3.40	4.58	5.43	6.02	6.06	5.34	4.30	2.80	1.77	1.19		
	RadStd		0.14	0.31	0.30	0.35	0.45	0.41	0.34	0.21	0.38	0.22	0.26	0.13		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	N/A

Nomenclature: See separate page